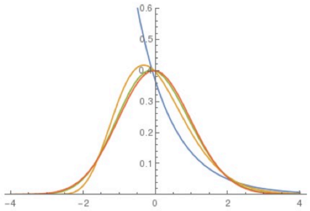
Exercise 3.1: Central Limit Theorem (CLT) (6P)

Let $X_1, X_2, \ldots, X_N$ be statistically independent and identically distributed random variables with the probability density function

$$p_X(x) = \begin{cases} e^{-x} & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$



With this exercise we would like to demonstrate the CLT for $Z_N = \sum_{i=1}^{N} X_i$ in the limit $N \to \infty$.

(a) Show that the N-fold convolution product of $p(x)$ is given by (1P)

$$p_{Z_N}(z) = p_X^{*N}(z) = \big(\underbrace{p_X * p_X * \cdots * p_X}_{N \text{ times}}\big)(z) = \frac{z^{N-1}}{(N-1)!} e^{-z} \qquad (z \geq 0).$$

(b) Check the norm and compute the mean and the variance of $p_{Z_N}(z)$. (1P)

(c) Standardize the distribution by shifting and rescaling Z_N in such a way that the new random variable $\tilde{Z}_N$ has a probability density $p_{\tilde{Z}_N}(z)$ with unit norm, zero mean, and unit variance (see figure). (1P)

(d) Show that the cumulant-generating function $K_{\tilde{Z}}(t)$ of the standardized probability density $p_{\tilde{Z}_N}(z)$ is given by (1P)

$$K_{\tilde{Z}}(t) = \frac{1}{2} N \ln N - t\sqrt{N} - N \ln\left(\sqrt{N} - t\right).$$

(e) Compute all cumulants κ_n and show that only κ_2 survives in the limit $N \to \infty$. What does it mean? (2P)

a) Compute some convolution products:

$$p_X^{*2}(z) = \int_{-\infty}^{\infty} e^{-x} \, e^{-(z-x)} \, dx =$$

$x \geq 0$
$z - x \geq 0$
$z \geq x$

$$= \int_0^z e^{-z} \, dx = z \cdot e^{-z} = \text{"} \frac{z^{2-1}}{(2-1)!} e^{-z} \text{"}$$

$$p_X^{*3}(z) = \int_0^z x \cdot e^{-x} e^{-(z-x)} \, dx$$

$$= e^{-z} \int_0^z x \, dx = \frac{1}{2} z^2 e^{-z} = \frac{z^{3-1}}{(3-1)!} e^{-z}$$

$$p_X^{*N+1}(z) = \int_0^z \frac{x^{N-1}}{(N-1)!} e^{-x} \, e^{-(z-x)} \, dx =$$

$$= e^{-z} \int_0^z \frac{x^{N-1}}{(N-1)!} \, dx = \frac{z^N}{N!} e^{-z}$$

$$= \frac{z^{(N+1)-1}}{((N+1)-1)!} e^{-z}$$

The formula seems to be valid for some examples and for the generalization for N

b) $$\int_0^\infty \frac{z^{N-1}}{(N-1)!} e^{-z} \, dz = \frac{1}{(N-1)!} \underbrace{\int_0^\infty z^{N-1} e^{-z} \, dz}_{\Gamma(N) = (N-1)!}$$

$$= \frac{(N-1)!}{(N-1)!} = \underline{\underline{1}} \rightarrow \text{normalization fulfilled}$$

$$E = \int_0^\infty z \cdot \frac{z^{N-1}}{(N-1)!} e^{-z} \, dz = \frac{1}{(N-1)!} \int_0^\infty z^N e^{-z} \, dz$$

$$= \frac{\Gamma(N+1)}{(N-1)!} = \frac{N!}{(N-1)!} = \underline{\underline{N}}$$

$$m_2 = \int_0^\infty z^2 \frac{z^{N-1}}{(N-1)!} e^{-z} \, dz = \frac{1}{(N-1)!} \Gamma(N+2) = \underline{\underline{(N+1) \cdot N}}$$

$$Var = E(z_N^2) - E(z_N)^2 = N$$

c) $$\tilde{z} = \frac{z_N - N}{\sqrt{N}} = \frac{z_N - E}{\sqrt{Var}}$$

* shift from N to origin
* reformation of the width with standard deviation $\sqrt{Var}$

d) $$M_{\tilde{z}_N} = E\left(e^{t \tilde{z}_N}\right) =$$

$$= E\left(e^{t \frac{z_N}{\sqrt{N}}} \cdot e^{-t \frac{N}{\sqrt{N}}}\right) =$$

$$= e^{-t\sqrt{N}} \underbrace{E\left(e^{\frac{t}{\sqrt{N}} z_N}\right)}$$

$$= e^{-t\sqrt{N}} \, M_{z_N}\left(\frac{t}{\sqrt{N}}\right)$$

$$K_{\tilde{z}_N} = \ln\left(M_{\tilde{z}_N}\right) = -t\sqrt{N} + \ln\left(M_{z_N}\left(\frac{t}{\sqrt{N}}\right)\right)$$

$$M_X(t) = E\left(e^{tX}\right) = \int_0^\infty e^{tx} e^{-x} \, dx =$$

$$= \int_0^\infty e^{-(1-t)x} \, dx = \left[\frac{e^{-(1-t)x}}{-(1-t)}\right]_0^\infty =$$

$t < 1$

$$= \frac{1}{1-t} \rightarrow M_{z_N} = \left(\frac{1}{1-t}\right)^N = (1-t)^{-N}$$

$$\Rightarrow M_{z_N}\left(\frac{t}{\sqrt{N}}\right) = \left(1 - \frac{t}{\sqrt{N}}\right)^{-N}$$

$$K_{\tilde{z}_N} = -t\sqrt{N} - N \ln\Big(\underbrace{1 - \frac{t}{\sqrt{N}}}_{\frac{\sqrt{N} - t}{\sqrt{N}}}\Big)$$

$$= -t\sqrt{N} - N \left(\ln(\sqrt{N} - t) - \ln\sqrt{N}\right)$$

$$= \frac{1}{2} N \ln(N) - t\sqrt{N} - N \ln(\sqrt{N} - t)$$

e) $$\kappa_n = \left.\frac{d^n K_{\tilde{z}_N}(t)}{dt^n}\right|_{t=0}$$

$$\kappa_1 = -\sqrt{N} - \frac{N}{\sqrt{N} - t} \cdot (-1) \Big|_{t=0} = 0$$

$$\kappa_2 = \frac{d}{dt} \frac{N}{\sqrt{N} - t}\Big|_{t=0} = \frac{-N \cdot (-1)}{(\sqrt{N} - t)^2}\Big|_{t=0} = \frac{N}{N} = 1$$

$$\kappa_3 = \frac{d}{dt} \frac{N}{(\sqrt{N} - t)^2}\Big|_{t=0} = \frac{-N \cdot 2(\sqrt{N} - t) \cdot (-1)}{(\sqrt{N} - t)^4}\Big|_{t=0} =$$

$$= \frac{2 N^{3/2}}{N^2} = 2 \frac{1}{\sqrt{N}} \rightarrow 0 \text{ for } N \to \infty$$

Equally for $n = 4; 5; \ldots$

For $N \to \infty$, we obtain $\kappa_2 = 1$ and $\kappa_n = 0$ for $n \neq 2$, which is characteristic for a gaussian distribution.

Physics of Complex Systems: Exercise 3

Leopold Schaller and Daniel Lender

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Task 3.1: Central Limit Theorem

(a)

(b)

(c)

see page 1

(d)

(e)

Task 3.2: Vector notation and Liouville operator

(a)

$$\dot{P}_c(t) = \sum_{c'} P_{c'}(t)\ \omega_{c'\to c} - P_c(t) \sum_{c''}\ \omega_{c\to c''} \tag{1}$$

to write the whole term as a sum over c', we use the Kronecker delta $\delta_{c',c}$. With $\delta_{c',c}P_c = \delta_{c',c}P_{c'}$ we get:

$$\dot{P}_c(t) = \sum_{c'} P_{c'}(t)\ \omega_{c'\to c} - \sum_{c'} \delta_{c',c}\Big(\sum_{c''}\ \omega_{c\to c''}\Big) P_{c'}(t) \tag{2}$$

$$= \sum_{c'} \Big(\omega_{c'\to c} - \delta_{c',c} \sum_{c''} \omega_{c\to c''}\Big) P_{c'}(t) \tag{3}$$

Now we expand at the Operator representation of the probability vector:

$$|P\rangle = \sum_{c'} P_{c'}(t)\, |c'\rangle \tag{4}$$

$$\Rightarrow\ \langle c|\dot{P}\rangle = -\langle c|\mathcal{L}|P\rangle = -\sum_{c'}\ \langle c|\mathcal{L}|c'\rangle\, P_{c'}(t) \tag{5}$$

$$= \dot{P}_c(t) \tag{6}$$

A comparison of both results shows:

$$-\langle c|\mathcal{L}|c'\rangle = \omega_{c'\to c} - \delta_{c',c} \sum_{c''} \omega_{c\to c''} \tag{7}$$

no we relabel the indices $(c \to c', c' \to c)$ and get the expression we should prove:

$$\langle c|\mathcal{L}|c'\rangle = -\omega_{c\to c'} - \delta_{c,c'} \sum_{c''} \omega_{c\to c''} \tag{8}$$

(b)

For the given transition rates, the Liouville Operator is:

$$\mathcal{L} = \begin{pmatrix} 3.5 & -1 & 0 \\ -3 & 3 & -3 \\ -0.5 & -2 & 3 \end{pmatrix} \tag{9}$$